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A big data-driven system for smart continuous improvement

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Abstract

Increasing digitalization and automation have elevated the importance of data analysis in reducing waste and costs, optimizing resources, and enhancing sustainability. However, as processes approach their optimization limits, traditional analytics become insufficient. This study introduces ImproveXpert4.0, a big data-driven platform designed to uncover actionable insights from process data through six algorithmic modules. These modules employ Machine Learning, Explainable AI, Operational Research, and Statistical methods for root cause analysis, bottleneck detection, simulation, prediction, human performance assessment, and variation prediction. ImproveXpert4.0 was developed using Design Science Research Methodology, and its algorithms were validated in real-world Manufacturing and Logistics settings. Its conceptual foundation is presented through two key artifacts: (1) the DMAIC-PDCA Structural Flowchart, which outlines its operating logic and algorithmic integration, and (2) a High-Level Architecture that details the technological infrastructure. Insights from a Narrative Literature Review and semi-structured interviews with Kaizen experts from award-winning organizations helped define the practical requirements for the system's design. The platform contributes to both theory and practice by promoting a structured, data-driven approach to Continuous Improvement in digital organizations.

Keywords Semi-Structured Interviews, Data-Driven System, Continuous Improvement, Root Cause Analysis, Performance Analysis, High-Level Architecture

1 Introduction and research background

In recent years, there has been a significant increase in the promotion and operationalization of digitalization in numerous countries, primarily promoted by public and private financing projects. Industry 4.0 and the data revolution have profoundly transformed companies, particularly in their industrial processes and human capital development [1]. These changes have been driven by integrating established technologies like the Internet-of-Things (IoT), Cyber-Physical Systems, Cloud Computing, and Big Data Analytics (BDA), which enable data generation, transmission, and application/analysis to enhance business efficiency, quality, and overall sustainability [2–4].



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While digitalization and automation have generally improved data-driven decision-making, they also introduce complex challenges that remain underexplored. For instance, traditional data platforms often struggle with the scalability and accuracy demands of Big Data [5]. The transition from traditional to agile methodologies in enterprise systems, such as the Enterprise Resource Planning (ERP) system, creates significant challenges in documentation and project management, including balancing lightweight and comprehensive documentation without sacrificing traceability or project outcomes. Furthermore, organizational resistance and resource gaps exacerbate the difficulties of adopting and integrating emerging technologies such as BDA [6, 7].

In addition, the preparation and orchestration of resources required for the actualization of BDA affordances remain poorly understood, creating gaps in how organizations can fully realize the potential of these technologies. There is also limited research into the role of BDA in complex and dynamic project environments, where its application for decision-making, risk management, and performance monitoring could significantly enhance outcomes [8]. These gaps point to a critical need for interdisciplinary approaches to develop frameworks that align technical capabilities with organizational readiness and cultural adaptation.

The benefits of digitalization are evident, particularly when combining industrial expertise with advanced data analytics to uncover patterns, trends, and root causes. Recent studies from Bernard et al. [9] and Fayyaz et al. [10], underscore the pivotal role of BDA in enhancing innovation, efficiency, and competitiveness across modern industrial sectors. Their findings highlight how BDA contributes to greater supply chain resilience and improved risk management in increasingly dynamic environments. For instance, digitalization in Portugal has driven a 43% productivity increase and a 34% rise in wages, with higher digitalization levels correlating with greater competitiveness and talent retention [11]. This success stems from embedding data-driven improvement initiatives into organizational practices, fostering a culture of Continuous Improvement.

Motivated by this potential and shaped by engagement with digitalization-focused industrial projects, this work proposes *ImproveXpert4.0*, a Big Data-driven system that provides industrial experts with insights comparable to those of commercial data analytics systems, but enhances them with innovative algorithmic modules drawn from academically validated methodologies that are still underutilized in practice. As the title suggests, *ImproveXpert4.0* aims to be among the first analytics platforms for CI that integrates Big Data, with a particular focus on the Volume and Velocity aspects of the four V's of Big Data. This integration entails incorporating technologies capable of handling such complex data.

ImproveXpert4.0 evolved organically from the initial goal of developing algorithms to address specific industrial challenges within national projects. It now serves as a data-driven solution for fostering Continuous Improvement, targeting Industry 4.0 organizations with sufficient digital maturity to ensure meaningful and practical application. *ImproveXpert4.0* is currently in an advanced prototyping stage and is being piloted using both real and simulated operational data from the Port of Sines in Portugal, as part of the NEXUS Agenda. A multidisciplinary research team, comprising mathematicians, data scientists, industrial engineers, and other relevant experts, is working on this prototype. The ongoing validation process is providing valuable feedback for refining the platform's algorithmic modules, integrating new ones, data integration pipelines, and

user interface to better align with the practical needs of Continuous Improvement teams operating in dynamic industrial settings.

The target audience for the ImproveXpert4.0 platform is Continuous Improvement teams/managers. Therefore, to help in the design and conceptualization, insights were gathered from experts in the area, so a Narrative Literature Review (NLR) and semi-structured interviews were conducted with ten Kaizen/Lean experts, six of them Kaizen-awarded company representatives. All insights were vital in identifying specific research activities necessary for the platform's design and further development, which are materialized in this work through the DMAIC-PDCA Structural Flowchart and the High-Level Architecture of the platform. These design elements are presented in a “language” tailored to showcase the platform's underlying functioning (the former) and enabling technological components (the latter), e.g., some of the technologies that handle Big Data and can be defined in the platform.

2 Research process

This section offers an overview of the entire research workflow, summarizing the steps and methodology utilized for designing/developing the platform and some of its incorporated algorithmic modules (see Fig. 1). Building on this, the research approaches

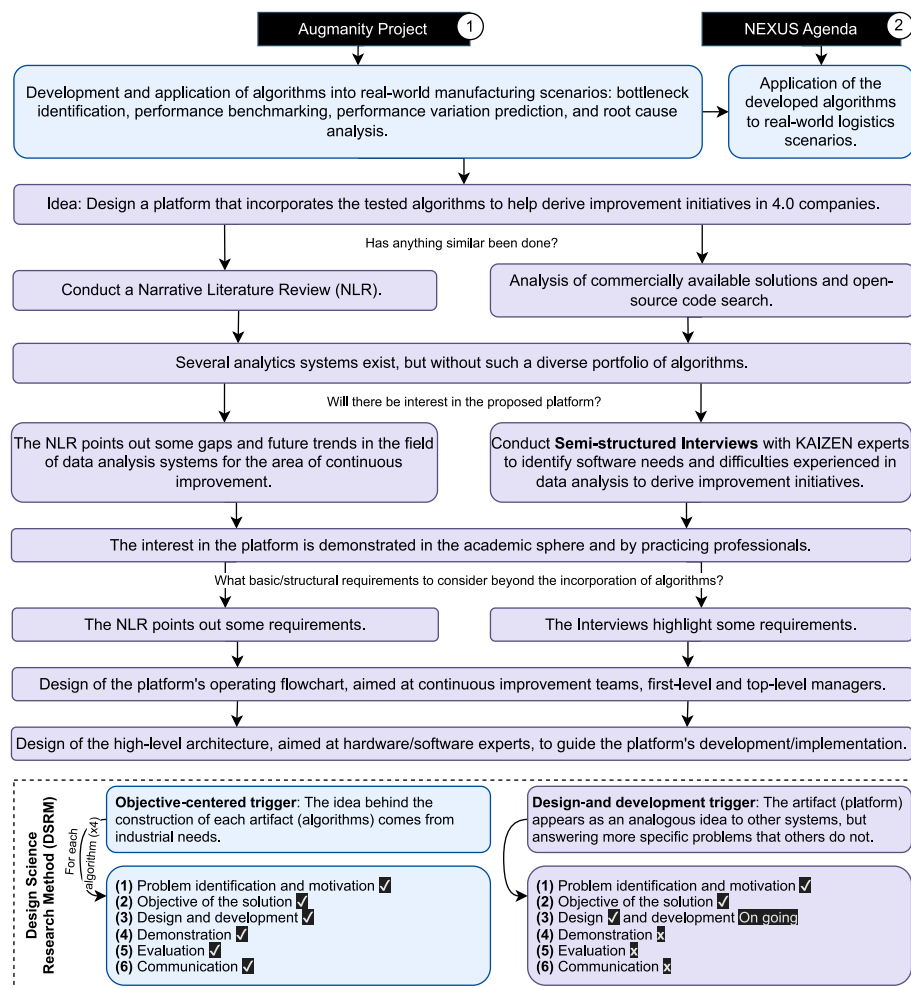


Fig. 1 Overview of the research workflow

employed to address the study's research questions and the development of respective artifacts are also outlined.

2.1 Development process overview

As illustrated in Fig. 1, this research was first motivated by real-world challenges introduced by industrial partners in Portuguese national projects: the Augmanity and the NEXUS Agenda. The first concerns the manufacturing domain, and the latter is regarding logistics and value chain management-related processes of a seaport. These challenges required the creation of innovative artifacts (algorithms) whose development and dissemination followed the Design Science Research Methodology (DSRM). Similarly, the ImproveXpert4.0 platform, as an information system artifact, followed the same methodology.

2.2 Design science research methodology

According to the definition of Hevner et al. [12], "Design science... creates and evaluates information technology artifacts intended to solve identified organizational problems" (p. 77). This methodology involves a structured process to design artifacts that address identified problems, contribute to research, undergo evaluation, and communicate results to relevant audiences. These artifacts, including constructs, models, methods, or implementations, encapsulate solutions to well-defined research problems [12].

Peppers et al. [13], leading authors on the DSRM, define six key activities: (1) Problem identification and motivation; (2) Definition of objectives for the solution; (3) Design and development; (4) Demonstration; (5) Evaluation; and (6) Communication.

Although presented sequentially, the DSRM allows flexibility, enabling researchers to start at any point in the process and iterate as needed [13]. Common entry points include:

- *Problem-centered approach (starting with activity 1)*: Triggered by observed problems or recommendations for future research;
- *Objective-centered approach (starting with activity 2)*: Driven by a research or industry need on a specific problem that requires a tailored artifact;
- *Design-and development-centered approach (starting with activity 3)*: When an existing artifact hasn't yet been applied to solve a specific problem. The artifact may come from another research domain or be an analogous concept;
- *Client-or context-driven approach (starting with activity 4)*: Arises from practical solutions that have been observed, often resulting from consulting experiences.

The algorithmic modules incorporated into ImproveXpert4.0 were primarily motivated by industrial challenges (Objective-centered approach), while the platform emerged during the design-and-development phase as an evolution of some data analytics systems, incorporating novel features enabled by the developed algorithms.

The general rationale behind the construct of both the algorithms and the platform is presented here for contextualization. However, this work is more focused on showing the details of the platform's design process, which hasn't yet been communicated. As Fig. 1 shows, all six steps of the DSRM were already conducted for the algorithms, while the platform is currently in stage three, within the development phase.

2.3 Research questions and solving approaches

To help guide the reader into the specific objective-centered and design-and-development triggers set for the algorithmic modules and the platform's design, Figure 2 displays the research ground of the entire work, namely its main research questions (RQs) and solving approaches/techniques.

The target audience of the ImproveXpert4.0 platform is Continuous Improvement teams and managers. To create a robust design for its structure, insights from the literature and industrial practitioners were gathered using both a narrative literature review and semi-structured interviews.

2.4 Data collection methods

2.4.1 Narrative literature review

In generic terms, the goal of an NLR is to examine and condense a body of literature, typically adopting a narrative, descriptive, and more generalist style to report the findings [14]. This is accomplished by providing an overview of the literature on the subject of interest, bringing attention to emerging research directions, and identifying areas that lack coverage [15]. Unlike systematic reviews, it does not follow a standardized search protocol, allowing for greater flexibility in selecting and synthesizing sources (some argue that it may lack reproducibility and methodological transparency) [16].

An initial search was conducted in Scopus using a combined query: TITLE-ABS-KEY ("big data" AND analytics AND technology AND business AND performance) AND TITLE-ABS-KEY ("managerial challenges" OR "technical challenges"). Based on the results, the search then focused on identifying and analysing highly cited research

DSRM trigger	Artifact	Research Question	Research Approach/Technique
Development and application of data algorithms according to industrial challenges proposed directly by companies	Bottleneck Detection Model	RQ1: How to detect, simulate and predict productive bottlenecks under the challenge of minimal information?	Narrative Literature Review
	Bottleneck Simulation Model		Queue-Directed Graph Modeling (problem representation, assumptions, state-space and bottleneck metrics)
	Bottleneck Prediction Model		
	Performance Benchmarking Model	RQ2: How to benchmark and assess overall performance or performance variation between process entities based on the principle of input consumption minimization and output production maximization?	Narrative Literature Review
	Root Cause Analysis Model	RQ3: How to identify the most relevant variables (root causes) that explain good/bad performance of process entities?	Multi-Directional Efficiency Analysis (MEA) Application using the overall performance-version (OP-MEA) or the performance variation-version (PV-MEA)
			Narrative Literature Review
			MEA Application using the OP-MEA or PV-MEA versions
	Performance Variation Prediction Model	RQ4: How can performance variations be predicted in a process operated by human workers?	Machine Learning Classification (XGBoost model included)
			eXplainable AI Techniques Computation (SHAP and LIME)
			Narrative Literature Review
			PV-MEA Application
Design of ImproveXpert4.0	ImproveXpert4.0	RQ5: How to design a fully data-driven platform (ImproveXpert4.0) that incorporates the above mentioned models and is appropriate/relevant for 4.0 Companies to derive novel Continuous Improvement initiatives?	Machine Learning Classification (XGBoost model included)
			eXplainable AI Techniques Computation (SHAP and LIME)
			Machine Learning Regression (XGBoost model included)
			Narrative Literature Review
			Semi-Structured Interviews

Fig. 2 Overall study's research questions and solving approaches/techniques

articles on the topics. A subsequent cross-citation analysis was performed to further explore the academic discourse. The primary objective was to uncover key challenges, limitations, and research gaps related to the application of big data analytics in driving continuous improvement and enhancing business performance in industrial companies.

As shown in Fig. 2, this review type was used during the development phase of the algorithmic modules, as well as in the design of ImproveXpert4.0. The primary objective for the latter was to highlight key concerns and considerations for its design, drawing on the set of challenges identified in the academic literature related to the digitalization journey. The results from this review can be found in Sect. 3.1.

2.4.2 *Semi-structured interviews*

After the NLR, ten semi-structured interviews were conducted to experts in Lean/Kaizen practice, aiming to achieve two primary objectives:

- Understand methodologies and challenges: Gain insight into the methodologies used by the experts in order to study industrial processes and develop improvement initiatives, highlighting the challenges encountered when utilizing data analytics systems;
- Identify platform requirements: Determine specific software functionalities and system requirements that should be incorporated into the proposed platform.

The first draft version of the interview questions was developed based on the knowledge acquired during the literature review and the practical experience gained in the development and application of the algorithmic modules presented in this work. In addition, domain experts were consulted to support the design of the interview questions, ensuring both linguistic clarity and content relevance. First, their input helped confirm that the language and terminology used were appropriate and easily understood by the intended participants. Second, they assisted in validating the content of the interview guide by assessing its alignment with field priorities, while also identifying potential gaps or redundancies in the draft questions. In addition, the interview plan also adhered to the detailed research protocol initially proposed by [17] and subsequently adapted by [18].

The interview guide, which contained six primary questions, was provided in advance to help participants prepare for the interviews. Taking into consideration the methodological approach outlined by [19], the questions were deliberately formulated at a broad and general level to reduce the influence of the interviewer's assumptions or perspectives and to minimize the risk of eliciting biased responses from the participants.

During the interviews, the original six questions were expanded to eight, addressing uncertainties and curiosities that arose during the discussions (please find the Interview Guide in this work's Appendix). These eight questions can be consolidated into four overarching broader questions (BQs):

(BQ1) How improvement projects emerge in companies, and which methodologies are used? - Questions 1, 2 and 3;

(BQ2) What is the most challenging phase of an improvement project, and what contributes to its complexity? - Questions 4 and 5;

(BQ3) What data types are used and analysed in these Kaizen projects? - Questions 6 and 7;

Table 1 Interview participants working for a particular company during the interview period

	ID	Company	Product	Recognition	Interviewee Job Title
Manufacturing	P1	BOSCH Thermo-technology Aveiro	Heating products and hot water solutions	<i>KAIZEN Award Portugal (2012), "Continuous Improvement system of excellence"</i>	Lean and Continuous Improvement Director
	P2	OLI S.A.	Cistern producer	<i>KAIZEN Award Portugal (2012), "Excellence on Productivity"</i>	Lean and Continuous Improvement Director
	P3	Vista Alegre S.A.	Luxury porcelain and crystal products	<i>Superior Excellence Award (2021) of Customer's Friend</i>	Quality Director
Services	P4	EDP Energias de Portugal	Electric utility	<i>KAIZEN Award Portugal (2018), "Excellence on Quality"</i>	Continuous Improvement Team Leader
	P5	Luis Simões S.A.	Transport and Logistics services	<i>KAIZEN Award Portugal (2016) "Excellence on Productivity"</i>	Lean and Continuous Improvement Director
	P6	BNP Paribas Portugal	Bank and financial services	<i>KAIZEN Award Portugal (2017), "Continuous Improvement system of excellence"</i>	Lean Six Sigma Team Leader

Table 2 Interview participants working for one or more companies during the interview period

	ID	Interviewee Job Title
Consulting	P7	Senior Lean Consultant and University Professor.
	P8	Co-founder of a Portuguese Kaizen Consulting Company and University Professor.
	P9	Portuguese President of a well-established non-lucrative Lean Association in Portugal.
	P10	Recognized Portuguese book author and Lean Six Sigma consultant.

(BQ4) How could emergent technologies enhance and expedite decision-making? - Question 8.

Participants were identified using a convenience sampling approach, leveraging personal connections and targeted searches through LinkedIn profiles. Before the interview, each expert was provided with comprehensive information about the study's objectives, procedures, and potential benefits, ensuring informed consent.

To assemble a diverse and qualified group, six company practitioners (P1–P6, Table 1) and four Lean/Kaizen consultants (P7–P10, Table 2) were selected based on the following criteria:

1. All participants should have a minimum of 5 years of work experience in Lean/Kaizen projects, serving as final decision-makers;
2. Company practitioners were chosen from organizations nationally or internationally awarded in Lean/Kaizen;
3. To diversify the range of perspectives, half of the company practitioners were working in manufacturing, while the other half in services (electric utility, transportation/logistics, and bank/financial services);
4. Consultants were selected based on both academic and practical experience in Lean/Kaizen implementation with recognized profiles in Portugal.

Detailed information regarding the interviewees' profiles is provided in Tables 1 and 2, with their identities kept confidential.

The interviews were conducted using videoconferencing platforms (Microsoft Teams and Zoom), as well as in-person. All participants provided consent for their interviews to be recorded exclusively for transcription purposes. Access to the recordings is restricted to the research team. Following transcription, the most notable responses and representative quotes were identified. The findings were meticulously analysed to highlight both similarities and differences across participants' perspectives.

3 ImproveXpert4.0: requirements and conceptualization

3.1 Insights from the narrative literature review

As mentioned before, this literature review sought to identify evidence supporting the necessity of the proposed data-driven platform by highlighting some key concerns and considerations that, according to the literature, should be addressed in its development. According to the literature, detecting and planning novel improvement initiatives driven purely by data remains a significant challenge for Continuous Improvement teams. Companies have faced various obstacles during their digital transformation journey, including:

- Technology Integration and Data Management
 - Aligning technologies with specific needs: Organizations often struggle to identify how new technologies can address their unique business challenges and integrate seamlessly into existing operations [20].
 - Merging data from diverse sources: Successfully consolidating data from systems with different characteristics (volume, variety, and velocity) poses challenges for achieving effective big data functionality [21–25].
- Transforming and Analysing Data
 - Extracting actionable insights: Converting raw data into meaningful information using AI techniques is inherently complex, exacerbated by a shortage of skilled professionals [26].
 - Ensuring data reliability and security: Organizations face difficulties in maintaining accurate, secure, and reliable data systems [27, 28].
- Designing and Managing Data Analytics Systems
 - Selecting and designing algorithms: Developing appropriate data algorithms is a challenging process [29]. Specifically, Machine Learning (ML) models often require extensive historical data for training, while root cause analysis (RCA) relies heavily on datasets with numerous variables (including categorical ones) and predefined expert relationships [30, 31].
 - Optimizing data exploration: Formulating effective questions to guide data analysis remains a complex task [32].
 - Visualizing data and reporting results: Designing user-friendly and informative visualization tools and result reports is often difficult [33, 34].
 - Creating user-friendly systems: Developing accessible data management systems that integrate algorithmic analysis, visualization, and exploration requires multidisciplinary expertise [33, 34].

- **Organizational Involvement:** Securing top management participation, i.e., gaining active engagement from senior leadership is often a significant challenge [33].

According to the literature, overcoming these challenges is paramount for companies. The goal is to acquire information and knowledge about industrial processes through the proper use of technologies, thereby enabling the design and planning of more effective Continuous Improvement initiatives within organizations.

Acknowledging the challenges mentioned above, it can be inferred that the following research activities are necessary for the design of ImproveXpert4.0:

1. The platform's basic operating structure must be designed, detailing the steps inherent to its logical operation. This structure should be grounded in a methodology familiar to companies, ensuring ease of understanding and acceptance. Alongside this, the platform's high-level architecture must also be developed to provide a comprehensive overview of its technological components, modules, and interactions. To ensure the proper design of this architecture, consultation with hardware and software experts is essential.
2. Incorporation of the performance variation prediction algorithm and the RCA algorithm described in [35–37]. These algorithms use efficiency analysis techniques, ML, and eXplainable Artificial Intelligence (XAI) methods to provide insights into performance and causality. Most importantly, the algorithms will not require large amounts of historical data or extensive variables for RCA, though the detail of the RCA may improve with the number of variables available.
3. Integration of a batch anonymization process to external company data that are injected into the platform's data warehouse and digital twin. This ensures data security and privacy while maintaining the functionality of the system.
4. To maximize the platform's impact and proper utilization, a training proposal must be developed to guide companies in using the integrated algorithms and formulating new problems for analysis based on their unique processes and data. Additionally, graphical requirements and essential characteristics should be collected to design an intuitive and effective user interface, ensuring a smooth user experience.

3.2 Insights from the interviews

The perspectives of academic experts on issues and concerns in the new technological era were pinpointed through the NLR. To complement this, the interviews aimed to gather insights into the methodologies and challenges encountered by professional practitioners during their work experience in the industry, as well as to explore how integrating some of the emerging Industry 4.0 technologies could improve decision-making within their Continuous Improvement projects.

Since the emergence of the term Industry 4.0 and its associated technologies in literature and industrial practices, there has been widespread recognition of their potential to enhance Continuous Improvement projects significantly. Such potential comes from extracting, treating, and exploring data from digital systems such as the ERP and the Manufacturing Execution System (MES) platforms, often enriched by IoT devices (e.g., in manufacturing shop floors) or, alternatively, from manual, non-digital tools that harken back to the origins of Kaizen and Lean. Since the present research focused on designing a purely data-driven platform to assist decision-making, interview participants

were explicitly instructed to confine their responses to projects that exclusively relied on data from digital systems or platforms.

In Question 1, participants were asked to describe how Continuous Improvement initiatives typically arise within companies - whether through a top-down approach, driven by top management KPIs (key performance indicators), cascading to operational-level metrics, or through a bottom-up approach originating at the operational level. The findings from the interviews revealed quite interesting insights. All manufacturing experts unanimously identified the top-down approach as the most influential in driving and encouraging improvement initiatives. However, service industry experts had distinct perspectives:

"The targets come from the top down. The improvement plans themselves come from the bottom up." (P4)

"It is 50|50. There is a central team that coordinates Continuous Improvement across the entire company. Then, there are local (operational) teams where improvement plans are defined more locally. Therefore, the aim is always to have a common area that cuts across the company, where improvement plans arise directly from a strategic plan, but from operational teams as well." (P5)

In contrast, three out of the four consultants noted that, in practice, the bottom-up approach is more commonly employed, even in organizations that perceive themselves as following a top-down strategy:

"Most problems or improvement projects arise from a bottom-up approach and not the other way around because from the top there is more of a vision, a direction that is intended to be achieved, where the improvement points are not very specific." P8

"The CEO or company manager is not always aware of the best ways to manage and lead, and often people on the ground decide to do this work, whether 1st, 2nd, or 3rd lines. Transformation attempts are more specific and conducted in pilot areas with people who know how to do so. Desirably, it should be top-down so there is a clear purpose in the implementation, someone who leads it and allocates resources with a sense of prioritization and responsibility. However, lean philosophy advocates that anyone, at any level, can drive this transformation." (P9)

Key insights from Question 1: Despite the differing perspectives among the experts, it can be concluded that Continuous Improvement projects naturally originate from both top management and operational teams. However, improvement targets are typically more clearly defined at higher levels of the decision-making hierarchy, while the design of improvement actions and plans tends to occur at the local level. Furthermore, the experts emphasized the significant influence of local improvement actions on higher-level KPIs, underscoring the importance of recognizing and explicitly defining this relationship to ensure the organizational integration of Continuous Improvement initiatives. Despite the different perspectives between experts, it can be inferred that Continuous Improvement projects naturally emerge from top management and operational teams. However, improvement targets are more clearly established at higher levels of the decision-making hierarchy, and improvement actions/plans are more carefully designed locally. Moreover, experts highlighted the impact that local improvement actions have on higher-level KPIs and how this relationship should be acknowledged

and specified to facilitate the organizational integration of Continuous Improvement projects.

Question 2 focused on the steps or methodological approaches companies employ to execute their improvement projects. Overall, the experts provided consistent responses, though their perspectives on implementation varied. Both the PDCA (Plan-Do-Check-Act) cycle and the DMAIC (Define-Measure-Analyse-Improve-Control) methodology were identified as the preferred approaches, with service industry experts exclusively referencing the use of the PDCA. Notable distinctions in their application were highlighted in the responses of P2 and P9:

“The general methodology for executing improvement projects is, without a doubt, DMAIC. (...) The DEFINE phase involves developing the process chart, writing the objectives, key rules for conducting the project, teams, deadlines, potential gains; in MEASURE, the measurement system is validated, the data are arranged to validate further whether we have improved or not; ANALYSE is analysing the root causes; in IMPROVE, actions with greater potential for improvement are implemented; and in CONTROL we check whether there has been an improvement and how the results are sustained and do not disappear after a while. The PDCA cycle enters the IMPROVE phase of DMAIC to validate the set of defined actions. If (the proposed actions) are not good, a new PDCA must be carried out. It works as an iterative method for conducting mini-projects to try to validate the initially suggested actions. However, the scope of the project does not change. Thus, DMAIC allows us to give a more global view, and PDCA is used for particular cases of mini-projects that belong to a larger global project.” (P2)

“I use PDCA from a more holistic perspective of managing improvement transformation. DMAIC appears to provide a deep dive into the data, where it is important to have statistical processing tools, to be able to isolate parameters or process variables that influence what is intended to be measured or improved.” (P9)

Question 3 focused on understanding how the outcomes of improvement projects are evaluated - specifically, how experts determine whether the implemented countermeasures led to real improvements. All experts provided similar responses, highlighting that such evaluations rely primarily on the analysis of quantitative KPIs established at the project's outset. Additionally, experts from service-based companies consistently emphasized the importance of formalizing these KPIs at an individual level, such as tracking metrics like hours saved per person or revenue generated per employee.

Key insights from Questions 2 and 3: Although expert opinions on the matter varied, the recommendations put forward by P2 regarding the most suitable methodological approaches for conducting Continuous Improvement projects have been adopted. P2 provided the most comprehensive and robust rationale, suggesting that the DMAIC protocol should be implemented at a macro level, while the PDCA cycle should be applied iteratively within the Improve phase of DMAIC, serving as a localized guideline. In alignment with the feedback from Question 3, operational or senior managers in any organization typically assess performance through the analysis of business metrics or KPIs. These metrics offer a tangible perspective on the company's operations, providing insights into their past, present, and, potentially, future (when predictive models are used). Therefore, it is logical to select DMAIC as the overarching Continuous

Improvement methodology, given its status as a well-established, data-driven framework. This choice places data (and, subsequently, information) at the core, serving as the fundamental assessment tool for any improvement initiative.

In Question 4, experts were asked regarding the stage(s) of Continuous Improvement projects that they considered to be the most challenging and time-consuming. Globally, experts agreed that the two most time-intensive phases were, first, RCA, and second, data cleaning, with the former being regarded as the most challenging. Furthermore, RCA garnered particular attention from respondents, with the duration of this phase varying widely, ranging from a minimum of two days to a maximum of three months. This variation in timespan was attributed to several factors, including:

1. The need to gather larger, specialized teams for brainstorming - cited by **P1**, **P6** and **P7**;
2. The requirement for further problem specification and the execution of mini PDCA projects from different problem angles - noted by **P3**, **P4**, and **P9**. Additionally, an interesting comment on the topic, *"(...) the greatest difficulty lies when dealing with a general problem, where there are N possible factors that justify it..."* (**P3**);
3. The collection of new data to analyse trends and fluctuating behaviors in the present, typically leading to project delays - as observed by **P2**, **P9**, and **P10**;
4. A belief in the team's capacity to identify the root cause and postpone the continuation of the project to a period when this analysis arises more naturally - mentioned by **P8**.

Subsequently, Question 5 was presented to participants P2, P4, P8, and P9 in response to the need to explore the insights they had provided in the previous question more deeply. The aim was to gain a better understanding of the methodologies, techniques, or tools the experts employed during RCA. The three most commonly used tools were the Pareto chart, the 5 Whys, and the Ishikawa diagram. Typically, Pareto chart analysis is performed in MS Excel. The 5 Whys and Ishikawa diagrams were manually constructed without the support of any computerized tools.

Question 6 focused on identifying the types or categories of data collected from digital systems and analysed by the experts during Continuous Improvement projects, in line with the initial condition provided to all participants. The three interviewees working in service companies reported utilizing worker performance data sourced from their company's ERP system. In contrast, all representatives from manufacturing companies stated that they relied on data extracted from both the company's ERP and MES systems, primarily concerning industrial processes conducted at workstations, such as machine-related metrics.

Out of curiosity, a supplementary Question 7 was posed to these manufacturing experts, inquiring whether they had ever considered worker-related data in their analyses during Continuous Improvement projects. Two respondents answered "No", without hesitation, while the third offered the following explanation:

"The objective is not to track the people but the process. More for ethical reasons."
(**P1**)

Additionally, two consultants, P8 and P10, indicated that they utilized both human and machine-related data in their projects. However, owing to their more extensive consulting experience with manufacturing companies, one of the consultants highlighted the following aspects: "I haven't used (worker-related data) because the use

of direct performance data or personal metrics about workers in the manufacturing industry is not well regarded ethically. However, it would be important to include such data because people are great sources of variability and the first actors in the production line. Their training, motivation, and health status, among others, directly affect the entire process. Additionally, there are often operational problems that any other reason cannot justify. Still, the cause is directly related to workers and their well-being in the workplace.” (P9)

Key insights from Questions 4 to 7: The majority of experts identified RCA as the most time-consuming and challenging phase of Continuous Improvement projects. Furthermore, experts typically rely on basic RCA techniques, such as the Pareto Chart, and do not have access to advanced computerized tools capable of identifying, for example, relationships between data variables to infer root cause dependencies. Regarding the data used, manufacturing experts predominantly focus on machinery (or non-worker-related) data, whereas worker performance data is considered essential by experts in service industries. Nevertheless, some manufacturing experts acknowledged the importance of incorporating human-related data. These observations suggest that, overall, experts lack access to more efficient and intelligent, data-driven tools or mechanisms for RCA, particularly when addressing multi-factorial issues. The integration of such a tool, capable of incorporating both human and non-human data (derived from work operations), could help account for all potential sources of process variation and facilitate a smoother transition to Industry 5.0. Following, in Question 8, participants were invited to share their perspectives on how emerging technologies could enhance and expedite decision-making, particularly in light of the previously discussed challenges related to RCA. Based on their responses, the experts proposed the following ideas:

1. A tool capable of identifying an improvement direction. Given the vast array of existing KPIs, such a tool would be able to determine, over a defined period, which process variables could yield the most significant performance improvements for the organization;
2. A tool that could provide insights into seasonality metrics, anomalous behaviors, or trends within the data, enabling the identification of key periods for study and intervention. More ambitiously, this tool could also suggest which potential causes might be most relevant to such events, along with their respective levels of importance;
3. A tool to facilitate data cleaning, ensuring that analyses and decision-making are not based on unreliable or erroneous data;
4. A tool capable of dynamically calculating theoretical KPIs, incorporating the experience and behavior of workers in their respective workspaces, to offer more accurate performance benchmarks and foster fairer performance evaluations.

4 ImproveXpert4.0: system design

4.1 DMAIC-PDCA structural flowchart

Based on the insights derived from the experts' responses to Questions 1–8 and the NLR, Figure 3 provides a summary table of the key findings and the associated research activities required to address them. Building upon these research activities, Figure 4 presents the DMAIC-PDCA Structural Flowchart, which represents the primary design

	Findings	Platform's Research Activity
Interviews	Question 1 Despite some different perspectives, continuous improvement projects naturally emerge from both top management and operational teams. However, improvement targets are more clearly established at higher decision-making hierarchy levels, and improvement actions/plans are more carefully designed locally. Moreover, experts highlighted the impact that local improvement actions produce in higher-level KPIs and how such a relationship should be acknowledged and specified towards organizational integration of continuous improvement projects.	Incorporate local performance metrics from the lowest decision-making levels (e.g., the shop floor) and global evaluation metrics from the top management level, ensuring that the former directly influences the global ones and that this impact is clearly understood.
	Questions 2 and 3 While expert opinions varied, P2's recommendations were highly considered. Providing the most comprehensive and robust explanation, P2 suggested that the DMAIC protocol should be employed at a macro level, and the PDCA cycle should be applied, as many times as necessary, within the Improve phase of the DMAIC approach, as a local guideline. Moreover, aligned with the feedback from Question 3, operational or top managers from any company evaluate performance by analyzing business metrics or KPIs. These metrics offer a tangible perspective on the company's operations, providing insights into their past, present, and, potentially, future (when predictive models are used). Consequently, it seems logical to select the DMAIC, a well-established data-oriented methodology, as the most overarching continuous improvement protocol followed in a company. This choice places data (and subsequently, information) at the core, serving as the fundamental assessment tool for any improvement initiative.	Make the DMAIC methodology the foundation for the platform's underlying functioning and use the PDCA as an iterative mechanism for improvement. The Define stage should encompass a comprehensive understanding of the company. The Measure phase must include robust mechanisms for data collection and processing. The Analyze stage should identify specific directions for improvement based on data variables. The Improve phase must integrate PDCA cycles according to the number of improvement actions required to achieve the desired performance goals, and the Control part needs to join all pieces together, providing a global picture of the impact of each PDCA on high-level metrics that affect the company at a macro level.
	Questions 4 to 7 Most experts pinpointed root cause analysis as the most time-consuming and challenging stage during continuous improvement projects, where sometimes new data collection endeavors are needed. Moreover, experts utilize simple root cause analysis techniques (e.g., the Pareto Chart) and do not have access to any intelligent computerized tool capable of defining, e.g., relationships between data variables to infer root cause dependency. Regarding data, manufacturing experts clearly preferred analyzing machinery (or non-worker-related) data, while worker performance data was vital for experts from service companies. These aspects demonstrate that experts lack the support of a faster and more intelligent data-driven tool or mechanism for root cause analysis when dealing with multi-factor problems. Such a tool could incorporate both human and non-human data (deriving from work operations) to accommodate all possible sources of process variation, allowing a smoother transition to Industry 5.0.	Incorporate the designed root cause analysis algorithm and allow the integration of human-related data exclusively for process performance measurement and analysis by management. Companies can choose to include this data based on their ethical policies. Also, a step in which the improvement project can be put on hold when data is missing.
	Question 8 Expert suggestions: (1) A tool capable of defining an improvement direction. Considering the vast amount of existing KPIs, such a tool should be capable of determining, for a certain period in time, which process variable(s) could give the biggest performance boost for the company; (2) A tool capable of providing information about seasonality metrics, abnormal behaviors, or trends in the data to find periods of study and intervention. Associated with it, and a little more ambitious, such a tool could also indicate which causes could be more relevant to such events and their respective percentages of importance; (3) A tool that facilitated data cleaning to avoid studying and analyzing a problem based on unreliable data; (4) A tool capable of dynamically calculating KPIs, taking into account the experience and behavior of workers in their workspace, to provide more reliable performance benchmarks and more fair performance evaluations.	Incorporate other relevant data algorithms that may encompass some suggested outputs (e.g., the bottleneck simulation and prediction algorithms). Introduce an extract, transform, and load step into the platform. Integrate the bottleneck detection algorithm that generates worker-related performance metrics, ensuring that this task proceeds concurrently with the previous one highlighted in Questions 4 to 7.
Narrative Literature Review	Overall difficulty in securing the participation of the top management (Dutta and Bose, 2015).	Integrate top-level management metrics into the platform and create a mechanism for evaluating them with every local improvement action to ensure their active relevance.
	Companies have difficulty understanding how new technologies can solve their business problems and integrate seamlessly with existing operations (Hizam-Hanafiah and Soomro, 2021).	Design the platform's basic operating structure, which contains the steps inherent to its operating logic, to ensure its functioning is clear and transparent for the end-user. Ideally, this structure should be based on a methodology familiar to companies to facilitate understanding and acceptance. In addition, design the platform's high-level architecture to provide a general view of the system's technological components, modules, and interactions.
	Combine data from diverse sources to achieve big data functionality (Bajic, Rikalovic, Suzic, and Pluri, 2021; Belhadi, Zkik, Cherrafi, Yusof, and El fezazi, 2019b; Gautham, Swaminathan, Shukla, Gaduparthi, and Malhotra, 2022; Lee, 2017; Zeid, Sundaram, Moghaddam, Kamarthi, and Marion, 2019).	
	Converting raw data into high-level information using AI techniques is complex. This is further compounded by a lack of skilled professionals in this area (Butt, 2020).	Incorporate the performance variation prediction algorithm and the root cause analysis algorithm, both of which combine efficiency analysis techniques, machine learning modeling, and explainable AI techniques. For these algorithms, there is no need to use substantial amounts of historical data or an extensive number of variables for root cause analysis. However, it seems evident that the level of detail of the root cause analysis may be affected by the number of variables used by the algorithm.
	Identifying and developing data algorithms is non-trivial (Saltz, Shamshurin, and Connors, 2017). Additionally, there are some constraints regarding the construct of these data algorithms, namely a prerequisite for extensive training of machine learning models using substantial amounts of historical data and for root cause analysis, a dependence on a substantial number of distinct variables within the dataset, including categorical variables, with a requirement for a priori expert-defined relationships among these variables (Ma, Li, and Thorstenson, 2021; Xu and Dang, 2020).	
	Cross-functional knowledge is required to design user-friendly data management systems (Dutta and Bose, 2015; Fadillah and Fitriana, 2019).	Consultation with hardware and software experts to support the design of the platform's High-Level Architecture.
	Companies face challenges in ensuring the accuracy and security of their information (Gupta, Modgil, and Gunasekaran, 2020; Moges, Dejegeer, Lemahieu, and Baesens, 2013).	Incorporate batch anonymization from company data sources into the platform's data warehouse and digital twin.
	Difficulties arise in designing appropriate questions to guide data analysis (Pandurang et al., 2023).	Proposal to train companies to use the algorithms integrated into the platform and to formulate new analysis problems based on their specific processes and available data.
	Designing relevant visual interfaces and result reports may be difficult (Dutta and Bose, 2015; Fadillah and Fitriana, 2019).	Gather graphical requirements and essential characteristics for the design of the platform's user interface.

Fig. 3 Summary of future research activities for the platform's development according to the interviews and the narrative literature review

output of the platform. This Flowchart is intended explicitly to describe the platform's internal components in detail, i.e., how it operates and performs in practice.

The DMAIC protocol represents a structured and well-defined framework that, in its initial stage (DEFINE), facilitates the integration of the organizational mission, vision, values, and purpose, alongside the requisite managerial analyses and KPIs for each company within a business group, particularly when the group comprises multiple companies.

After identifying areas for improvement within each company through the definition of relevant case studies, managerial teams further elaborate on each case. This process involves the identification and characterization of all available and pertinent data

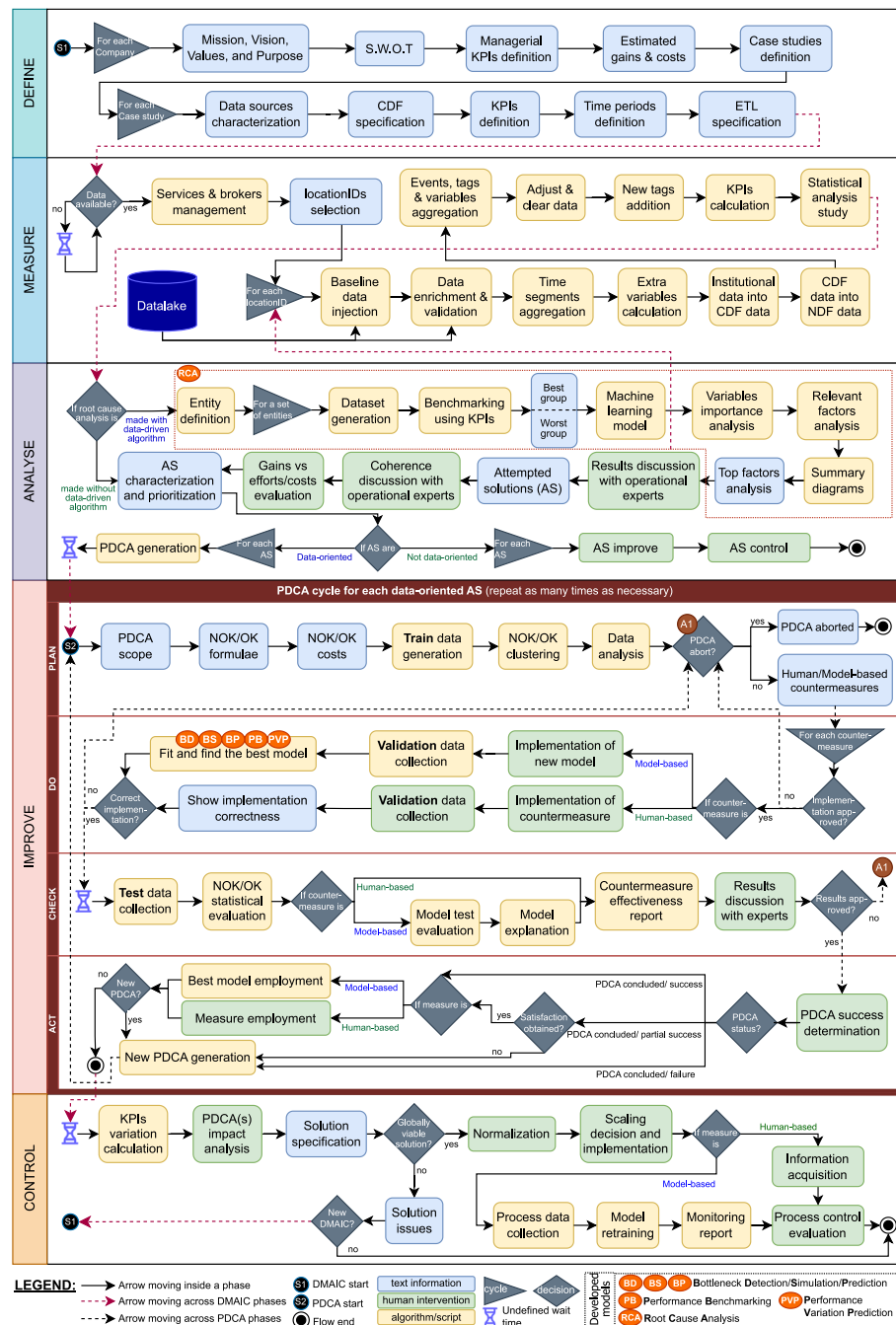


Fig. 4 The ImproveXpert4.0 platform: DMAIC-PDCA structural flowchart

sources, the specification of the CDF (Common Data Format),¹ the definition of the specific set of KPIs, and the determination of the time periods during which these metrics will be assessed. The process concludes with the specification of the ETL (Extract, Transform, Load)² process. It is important to note that all this text information is gathered

¹ According to [38], the CDF is defined as a “self-describing data format for the storage of scalar and multidimensional data in a platform- and discipline-independent way.”

² The ETL process involves consolidating data from various sources into a central repository, a data warehouse. It utilizes predefined rules to clean, organize, and prepare raw data for storage, ML, and analytics [39].

inside the platform. Additionally, all blue boxes represent text information that has been introduced into the platform.

In the MEASURE phase, the decision-maker assesses whether all necessary data for analysing a case study are available. If the decision is NO, the improvement project is placed on hold for an undefined period. During this waiting time, for instance, parallel initiatives, such as the deployment of IoT sensorization, may be pursued. Consequently, the MEASURE phase serves as a critical foundation for the platform. Based on the previously defined ETL specifications, this phase provides a systematic approach to performing the ETL processes, ensuring that the data is appropriately prepared for exploration and analysis in subsequent stages.

Once the availability of all relevant data has been confirmed, this stage is initiated by the management of services and brokers responsible for integrating and modeling streaming data. Subsequently, the set of *locationIDs* associated with the case study is defined, where these *locationIDs* correspond to the identifiers of the specific locations under study. For instance, if a case study is defined as “throughput improvement of the factory’s manufacturing lines,” the *locationIDs* would refer to the unique identifiers of each production line within the factory. A series of eleven scripts is then executed for each *locationID*.

The first two scripts connect with the company’s data lake,³ primarily to collect and inject baseline data associated with a given *locationID* into the platform. This process is complemented by enriching the baseline dataset with additional data, followed by a validation procedure.

The third script consolidates and aggregates all preceding data by time segments, establishing a defined temporal unit. Subsequently, additional variables may be computed in the next script. Following this step, institutional data undergoes a format conversion to the CDF, as per the specifications outlined in the Define stage. The CDF is then transformed into the Normalized Data Format (NDF), which serves as the operational data format for the platform.

After data normalization, the subsequent script aggregates events, tags, and quantitative variables. These data elements are then adjusted and refined in accordance with the ETL specification.

The final three scripts perform the following functions: first, they add any new tags required for analysing a specific *locationID*; second, they calculate the relevant KPIs based on definitions established during the preceding DMAIC stage; and third, they conduct a statistical analysis of the calculated KPIs, segmented by time. This statistical analysis is the foundation for the ANALYSE stage, which is introduced by a subsequent decision point. Please note that the platform itself is designed with the assumption that it has full access to all company data related to the specified case studies. In this framework, the Measure phase involves gathering all available data and calculating KPIs and metrics, without pre-selecting which ones to actually evaluate, in order to conduct the improvement initiative. The relevant variables are then identified during the ANALYSE phase, guided by the algorithmic support, to ensure that only the most pertinent ones, i.e., those that provide the highest improvement potential, are selected.

³ A data lake is a centralized repository designed to ingest and store vast amounts of data in its raw form. Its open and scalable architecture enables it to accommodate data from diverse sources, including structured data such as database tables and Excel files, semi-structured data like XML files and web pages, and unstructured data such as images, audio files, and social media content, while preserving data integrity [40].

The initial statistical insights provided during the MEASURE phase are key for decision-makers in determining whether the assistance of an advanced data-driven algorithm is necessary to guide them toward more specific improvement directions within a case study. If the decision-maker opts for the algorithm-driven approach, by the end of its application to a particular process, he is presented with the top most relevant factors that explain performance-related issues, i.e., a RCA for bad/good performance.

Conversely, if this RCA is conducted without utilizing the data-driven algorithm, specific improvement directions would have already been predetermined during the ANALYSE phase. In this scenario, decision-makers and operational experts collaboratively define the most cost-effective set of “possible solutions,” referred to in the Flowchart as attempted solutions (AS), to address the identified improvement areas.

Furthermore, given that the primary objective of the Flowchart - and by extension, the platform - is to deliver data-driven assistance, the decision-maker is required to specify whether each AS is data-oriented. This distinction determines whether the effectiveness of its implementation can be evaluated using data. Each AS is thus characterized and prioritized in a text file. Only those AS deemed data-oriented, progress to the subsequent DMAIC phase as a PDCA mini-project.

The IMPROVE stage starts with the application of the PDCA cycle to each data-oriented AS. Within the cycle, two critical steps emerge at the PLAN stage: (S1) the definition of a NOK/OK formula and (S2) the establishment of human- or model-based countermeasures. Step S1 involves specifying a unique quantitative evaluation metric for the PDCA initiative through an NOK (Not OK) or OK reference formula. For instance, $\text{NOK} = \text{throughput} \leq 50$ units per workstation. Step S2 determines whether the countermeasures take the form of mathematical or ML models or are direct human interventions that can be monitored and evaluated using process data.

To clarify, the primary distinction between data-oriented and non-data-oriented AS lies in the direct relationship between the collected data and the implemented AS. Many situations arise in which managers implement improvement solutions. However, the real impact of such a measure alone in the business process is not genuinely tangible because it affects a group of process variables that have a subjective nature associated with them.

At the DO stage of the PDCA cycle, there is a reference to five models (see step *Fit and find the best model* in the platform's Flowchart of Fig. 4): (M1) a bottleneck detection (BD) model, (M2) a bottleneck simulation (BS) model, (M3) a bottleneck prediction (BP) model, (M4) a performance benchmarking (PB) model and (M5) a performance variation prediction (PVP) model. A more detailed explanation of these embedded algorithmic modules can be found in the following subsection.

In the CONTROL stage, the objective is to evaluate “the impact (...) that local improvement actions produce in higher-level KPIs,” as emphasized during interviews. Following the completion of all PDCA implementations (indicated by the hourglass icon in Fig. 4), a platform script calculates the variation in the KPIs associated with a specific *locationID* within the context of a *case study*.

Subsequently, decision-makers assess the effects of the implemented PDCA cycles and document the final solution in textual form. With the final solution defined, they evaluate its global viability by determining whether the overall performance variation is positive or negative. This step is crucial, as effective local improvement initiatives (achieved through successful PDCA cycles) can potentially have adverse effects

on global performance. If such negative impacts are observed, the solution's issues are reported, and decision-makers must decide whether to initiate a new DMAIC protocol or abandon the solution altogether.

If the solution is deemed globally viable, experts proceed to normalize and prepare it for scaling and implementation across other companies within the business group. Depending on the nature of the measures outlined in the final solution, additional actions may be required: model-based measures will necessitate retraining after new data collection, followed by generating a monitoring report via a platform script. For human-based measures, their ongoing status is visually tracked. The DMAIC ends with a process control evaluation made by the experts.

For a practical and straightforward walkthrough of the five DMAIC stages applied to a real-life example, refer to Fig. 5.

4.2 Embedded algorithmic modules

To elucidate the macroscopic positioning of the algorithmic solutions incorporated into the DMAIC-PDCA Structural Flowchart, see Fig. 6. Each algorithmic solution is succinctly described in the subsequent subsections to offer a general overview of its objectives and intended purpose. Because they have already been tested and scientifically validated, information about the datasets used is detailed in their respective papers.

DEFINE	- Company X has a managerial meeting and defines 2 major improvement case studies. - One of the case studies refers to throughput improvement of the factory's manufacturing lines.	
MEASURE	- The case study locationIDs are: line1, line2, line3, line4 and line5. - For each locationID, KPIs are calculated and a statistical analysis is made, assessing global improvement targets.	
ANALYSE	- Decision-makers in collaboration with operational experts perform root cause analysis without using a data-driven algorithm - recall the techniques mentioned in the interviews to Question 5. - To improve the throughput of the factory's manufacturing lines, decision-makers established two prioritized AS to be employed in line2 as reference: (AS1) put new chairs in workstations operated by workers and (AS2) real-time identification of problematic workstations. - AS1 is not data-oriented, i.e., the company doesn't have the necessary data to effectively control and assess its impact. Hence, the new set of chairs is bought and placed in the manufacturing line and managers control if workers are using them correctly, providing basic maintenance instructions (steps AS improve and AS control). AS1 does not move forward within the DMAIC protocol due to its not data-oriented nature.	
IMPROVE	PDCA1 AS2	
	PLAN	PDCA scope: real-time identification of problematic workstations. NOK formula: workstation with an average global processing and queue time > 105% of the mean value. NOK costs: loss estimation of 10 manufactured products/month (approx. 3400€/month loss). Model-based countermeasures: (MC1) implementation of a bottleneck detection model and (MC2) implementation of a bottleneck simulation model.
	DO, CHECK and ACT	For MC1 and MC2, the phases DO and CHECK are followed systematically and successfully in the platform. In the ACT phase, the decision-maker determines: <i>PDCA concluded with success</i>, but decides to generate a new PDCA to address some of the issues identified by the bottleneck identification and simulation models.
	PDCA2 derived from AS2 implementation	
IMPROVE	PLAN	PDCA scope: improve the throughput of workstation3. NOK formula: throughput > 18 processed parts/work shift NOK costs: loss estimation of 10 manufactured products/month (approx. 3400€/month loss). Human-based countermeasure: (HC1) replace the motor of the workstation's machine for a new/faster one.
	DO, CHECK and ACT	For HC1, both phases DO and CHECK are followed systematically and successfully in the platform. In the ACT phase, the decision-maker determines: <i>PDCA concluded with success</i>, without the need to generate a new cycle.
CONTROL	- After calculation in the platform, decision-makers determine a positive KPIs variation resulting from the successful implementation of the previous PDCA mini-projects. Moreover, managers choose AS2 for implementation in other manufacturing lines of the company. The set of measures currently employed in AS1 and AS2 are periodically controlled by the experts.	

Fig. 5 Practical example of the platform's flowchart functioning

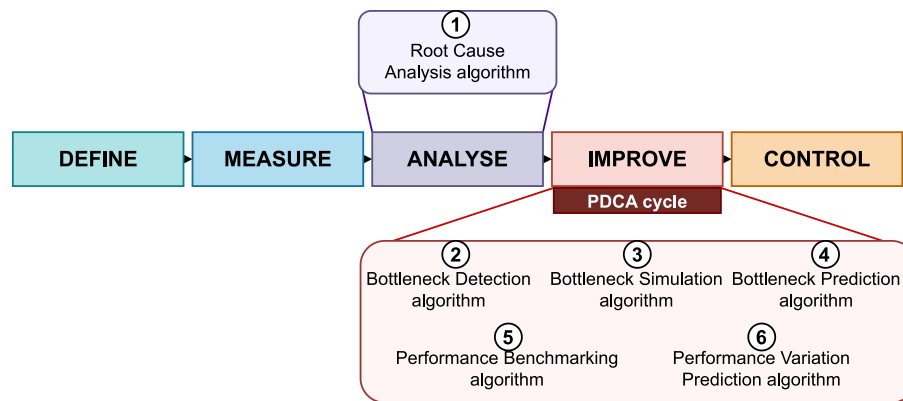


Fig. 6 Location of the algorithmic models within the DMAIC-PDCA structure

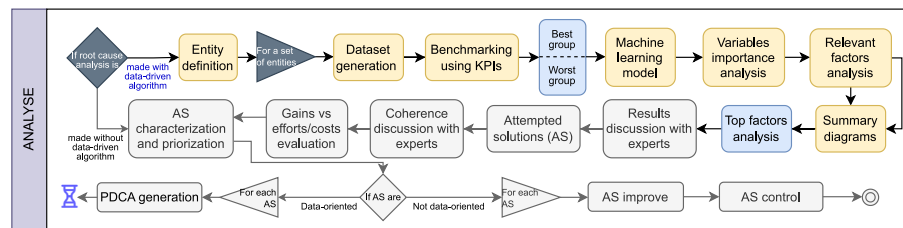


Fig. 7 Root cause analysis algorithm flowchart

4.2.1 Root cause analysis

At the beginning of the DMAIC's Analyse phase, the decision-maker can decide between using (or not) a data-driven algorithm to perform RCA, i.e., to identify an improvement direction. Suppose he decides on the algorithm path by the end of its application to a particular process. In that case, he is presented with the most relevant factors explaining performance-related issues. To highlight the steps employed by the data-driven RCA model, please attempt the colored Flowchart from Fig. 7.

The novelty of the designed, tested, and validated RCA algorithm comes from the combination of well-known techniques derived from Operational Research, ML, and XAI (see [35–37]). Looking at Fig. 7, after the decision point, the decision-maker starts by defining the so-called *Entity*, i.e., the decision-making unit that will be evaluated. Taking one of the case studies analysed as a reference, an Entity may be represented, e.g., as a tuple in the form of (*workShiftID*, *workerID*), representing workers who may be operating in different work shifts. The RCA algorithm initiates after generating the dataset for the predefined set of entities. In both case studies explored by the authors, relevant process KPIs are consumed and produced by the entities, which are then benchmarked/ranked using Multi-directional Efficiency Analysis (MEA), an advanced technique derived from Data-Envelopment Analysis (DEA). In what follows, the benchmarked entities are split into two groups based on a threshold and used as input for an ML model. The data-driven RCA algorithm concludes by analysing the top relevant factors computed through XAI techniques, e.g., using SHapley Additive exPlanations (SHAP). For a better understanding of the intricate details of the algorithmic solution, see the works of [35–37].

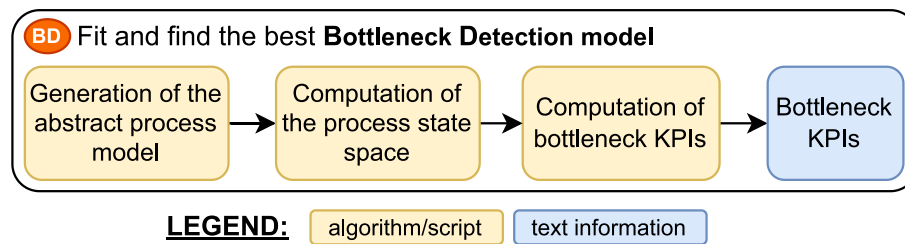


Fig. 8 Bottleneck detection algorithm flowchart

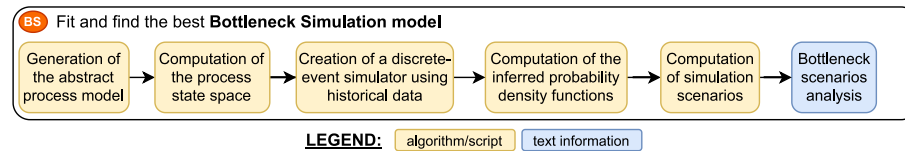


Fig. 9 Bottleneck simulation algorithm flowchart

4.2.2 Bottleneck detection

One of the first algorithmic solutions developed was a BD model whose objective was to identify which stages, processes, or resources were limiting the capacity of a system, causing delays or inefficiencies. The area of BD has proven to be relevant in many fields, such as logistics, manufacturing, and information processing. The proposed solution was tested and validated in an industrial manufacturing case study and based on the concept of *minimal information* defined by [41]. According to the authors, minimal information represents the lowest level of information that describes some system (e.g., for a manufacturing line, such data are their *workstation IDs*, the *part IDs* - unique references of the parts under processing in the production line, and the *workstation exit timestamps*). In the case study analysed by [41], the authors used a Queue-Directed Graph (QDG) model to represent the manufacturing line abstractly. Then, through a specific mathematical state-space of the problem, the *minimal information* extracted from the company's MES was enough to autonomously compute relevant BD KPIs (based on queue state, process state, and a combination of both).

Although the BD model was primarily tested in a manufacturing process context, any system that can be modeled as a QDG may benefit from it, enabling the identification of elements that negatively influence the system's performance. To apply the BD model, Fig. 8 shows a simple Flowchart of the algorithm behind the model. For further enlightenment, please see the work [41].

4.2.3 Bottleneck simulation

In alignment with the previous algorithm, the BS model proposed in [42] was also integrated into the platform (see Fig. 9). It consists of a discrete-event simulator based on the Python Package Salabim 21.0.4 (<https://www.salabim.org/manual/>), where experimental distributions are modeled from historical data. In the case study explored by the authors, the BS model was employed to understand how a manufacturing system behaves under different operational scenarios, allowing the exploration of pertinent 'what if' questions without experimenting with the system itself. This constitutes a significant advantage of the BS model usage and the reason why it provides important complementary insights

for decision-making. Similar to the previous algorithm, the discrete-event actions were ruled by a formal model also defined as a QDG.

4.2.4 Bottleneck prediction

While the BD model allows historical and current analysis, and the BS model studies “parallel” scenarios of bottleneck occurrences, this final bottleneck-related algorithm is directed to the future. The BP model aims to devise an early warning system that triggers decision-makers to act within a reasonable timeframe in their production system (see Fig. 10). The BP model included in the ImproveXpert4.0 platform complements the previous two models. It also follows a formal model defined as a QDG, allowing its application and employment in different manufacturing, services, or logistics systems where such a formal structure is respected. For more detail about the model proposal and case study application, please see the work from [42].

4.2.5 (Human) Performance benchmarking

The developed (Human) PB model emerged from a problem identified in a manufacturing case study [43]. The initial goal was to create a tool to help managers make decisions with less subjectivity regarding their workers’ (human component) evaluation scheme, i.e., using performance KPIs calculated from data extracted from the company’s MES. Thus, a PB model seemed the most appropriate one as it provides a ranking or *efficiency score* based on the idea of input consumption-output production. For clarity, taking the case study analysed as the reference, a set of workers operating in work shifts were evaluated based on their job *experience time* and *wage* (inputs), the job *delay time* and their *response time* to new parts in a manufacturing line (outputs).

The benchmarking technique chosen for the case study application was MEA, whose great advantage compared to traditional DEA is that it can incorporate situations where, e.g., by doubling inputs, outputs are not directly doubled too. Thus, the benchmarking algorithm identified individual inefficiencies per worker operating in a work shift and clustered them according to similar inefficiency parameters (please see the Flowchart from Fig. 11).

Adding to the more human-related manufacturing industry application, the model has also been tested in three logistics case studies. In the work from Brochado et al. [44], the PB model was employed to evaluate multimodal transportation routes of containers managed by distinct logistics service providers, in a subsequent paper from the same main author [37], the performance of last-mile delivery services was benchmarked, and a more recent work from Kazmi et al. [45], makes an assessment on performance of port equipment, such as cranes and loading arms. These applications demonstrate the relevance and adaptability of the PB model to various distinct contexts and problems.

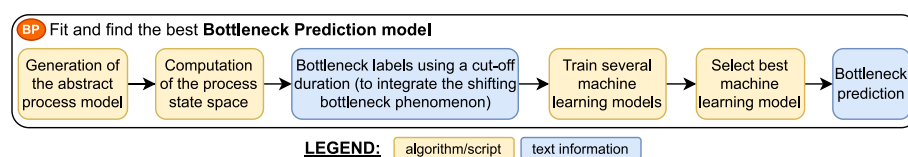


Fig. 10 Bottleneck prediction algorithm flowchart

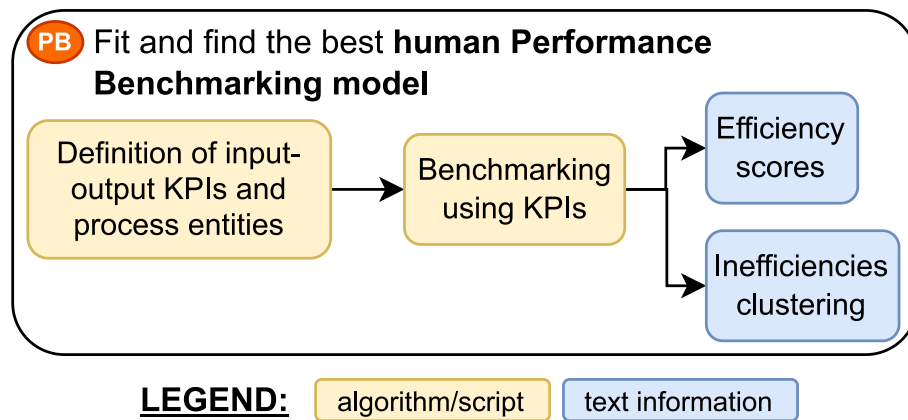


Fig. 11 Performance benchmarking algorithm flowchart

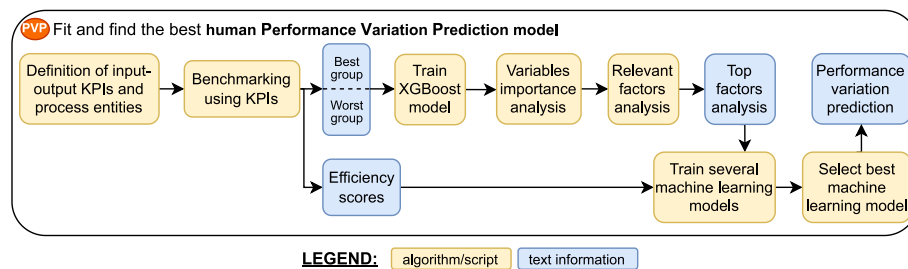


Fig. 12 Performance variation prediction algorithm flowchart

4.2.6 (Human) Performance variation prediction

The last algorithmic solution represents a (Human) PVP model. The human component comes from the fact that the approach was tested and employed in two case study scenarios with manufacturing line workers [35, 36] and further validated by their managers. As the name suggests, the predictive model is founded on data related to performance variation, i.e., based on the current performance status of a set of entities. Following the Flowchart in Fig. 12, these data are obtained using the benchmarking and XAI approaches described in Sect. 4.2.1. However, the differentiating factor in analysing the variation in human performance is the concept of *transition of consecutive periods*, where the set of input/output KPIs benchmarked per entity is first assessed on a time basis.

This data-driven approach has proven valuable for managers in identifying and analysing the periods when their workers showed significant performance drops.

4.3 High-level architecture

The high-level architecture of ImproveXpert4.0 encompasses the system's general framework, presented conceptually in Fig. 13. This architecture provides a comprehensive view of the system's technological components, modules, and interactions. It facilitates an in-depth understanding of its internal structure for software and hardware specialists, aiding in practical implementation. Simultaneously, it offers decision-makers a clear perspective on the system's technological structure and capabilities.

The ImproveXpert4.0 web interface features a controller that can integrate data from five distinct external sources. Given the variability in data characteristics across

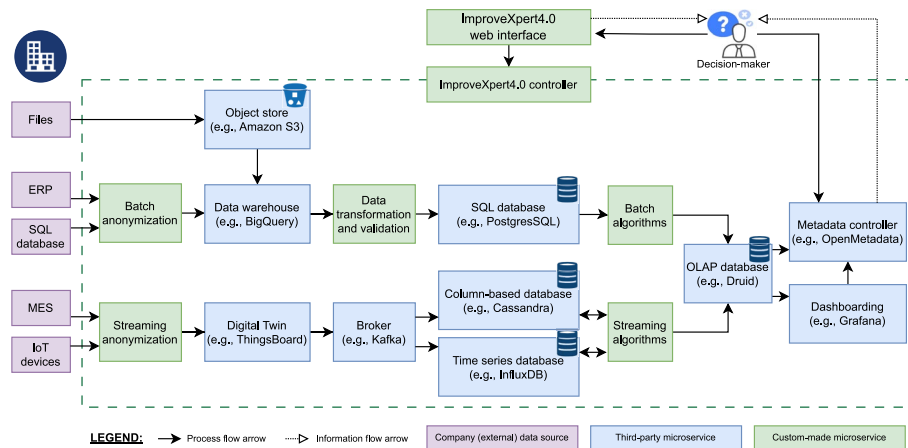


Fig. 13 ImproveXpert4.0: high-level architecture

companies - each exhibiting unique features and particularities - specific stages within the primary controller microservice need custom-built microservices to address these variances effectively (e.g., batch or streaming anonymization). Through the platform, decision-makers can actively interact with the ImproveXpert4.0 web interface and the Metadata Controller, giving inputs and extracting knowledge from the data insights.

Overall, the ImproveXpert4.0 platform offers capabilities of (1) metadata discovery, (2) data collaboration for enhanced understanding between decision-makers, (3) data quality by instilling trust in data through quality tests that ensure freshness and accuracy, (4) data lineage, i.e., allowing the path of data to be traced across tables, pipelines, and dashboards, (5) data insights, through KPIs definition, (6) data governance strengthening, and (7) advanced data analytics using integrated algorithmic solutions. Collectively, these features highlight the platform's significant benefits and scalability potential, making it a robust tool for modern data-driven environments.

5 Conclusions

In the face of increasing digitalization, traditional analytics methods are proving inadequate for uncovering improvement opportunities in complex, data-rich environments. Existing Continuous Improvement platforms often lack algorithmic diversity and fail to integrate established methodologies like DMAIC and PDCA, resulting in fragmented, narrowly focused solutions. Companies also struggle to manage and merge heterogeneous data sources from ERP, MES, and IoT systems, while maintaining data quality, privacy, and usability. Furthermore, many platforms overlook the impact of human factors on process variation, particularly in manufacturing, and lack tools to ethically and effectively incorporate such data. Root Cause Analysis remains a particularly challenging and time-consuming task, often performed manually with limited analytical depth. Additionally, organizations face difficulty linking localized improvement actions to global KPIs, which impedes strategic alignment.

In light of these research gaps, this work proposes ImproveXpert4.0, a big data-driven platform that integrates a suite of six innovative algorithmic modules to derive non-trivial and unconventional insights. Those modules combine Machine Learning, eXplainable AI, Operational Research, and Statistical techniques for: (1) root cause analysis, (2)

bottleneck detection, (3) bottleneck simulation, (4) bottleneck prediction, (5) (human) performance benchmarking, and (6) (human) performance variation prediction.

The platform's design and the development of its algorithmic modules were guided by the well-known DSRM methodology. While the algorithms have already been tested with real case applications in Manufacturing and Logistics and disseminated in the academic literature, the conceptualization and design of ImproveXpert4.0, articulated through its DMAIC-PDCA Structural Flowchart and High-Level Architecture, have not. That is the contribution of this research work. The DMAIC-PDCA Structural Flowchart encapsulates the operational principles of the platform and its integrated algorithmic modules. At the same time, the High-Level Architecture delineates the necessary technologies and interconnections for its implementation. To meet current technological requirements and understand overall considerations regarding the platform's operating structure, this work was supported by an NLR and semi-structured interviews conducted to Kaizen/Lean experts in different business areas, most of them working on Kaizen-awarded companies. The combination of both research approaches enabled the identification of several essential research activities related to the platform's concept design, emphasizing their importance to the outcomes of this study.

It is important to note that the integrated algorithms are referred to as “algorithmic modules,” as they are designed to function as plug-ins - modular components that can be selectively incorporated into or excluded from the platform based on a company's specific data analysis requirements.

5.1 Theoretical implications

The modules incorporated in ImproveXpert4.0 contribute to advancing theories in Kaizen/Continuous Improvement, particularly in Manufacturing and Logistics, by prioritizing data analytics in operational decision-making. This work highlights the theoretical importance of leveraging real-time and historical data to enhance industrial processes by emphasizing data as a core component of the Continuous Improvement process.

- The *Bottleneck Detection, Simulation and Prediction models* provide a structured, minimal-data approach for identifying, simulating, and predicting bottlenecks based on a QDG representation of the process. To the authors' knowledge, this theoretical model is the first in the literature able to compute bottleneck metrics using only three main variables, the *partNumberID* (tokenID), the *locationID* (nodeID) and *departureTime* (timestamp of a token leaving a node);
- The *Process Performance Analysis model* enhances the understanding of efficiency by distinguishing between the overall (fixed period) and the variation-oriented version of performance analysis. This further expands the theoretical discourse on performance measurement by highlighting areas where efficiency variation, rather than “static” efficiency, is meaningful to evaluate operational processes;
- The *RCA model* combines XGBoost with well-known XAI techniques like SHAP/Lime for purely data-driven causality analysis. Moreover, the model's ability to clarify the contribution of specific variables in efficiency data about Manufacturing or Logistics processes speaks to a theoretical shift towards more transparent data-driven operation decisions;
- The *Human (Workers) Performance Variation Prediction model* introduces the human element to predictive modeling, aligning with Industry 5.0 principles, where

worker-centric data are acknowledged as integral to process optimization. This model expands theoretical frameworks to incorporate the human factor in production;

- The *DMAIC-PDCA Structural Flowchart* brings forth an integrated representation combining the DMAIC with the PDCA methodology, theorizing an improved framework for Continuous Improvement in Industry 4.0. This framework could be a comprehensive example for other researchers and practitioners in the field. Moreover, it represents the theoretical foundation for a big data-driven platform, the *ImproveXpert4.0*.

5.2 Managerial implications

The developed models and platform equip companies with streamlined solutions to enable sustained Continuous Improvement practices, with a strong focus on enhancing organizational performance.

- The *Bottleneck Detection, Simulation, and Prediction models* offer managers a simplified yet effective tool for pinpointing, simulating, and predicting process bottlenecks, allowing for timely interventions with minimal data requirements. Companies, particularly in resource-constrained environments (where IoT implementation is still in its infancy), can identify process bottlenecks without extensive data collection efforts, reducing both operational and data processing costs.
- The *Process Performance Analysis model* enables managers to compare performance through two lenses: overall efficiency and variation-focused efficiency. This provides nuanced insights into operational performance as a whole and throughout time, allowing managers to analyse efficiency drops and areas of varying efficiency that may signal instability in Manufacturing or Logistics processes.
- The *Root Cause Analysis model* empowers managers to diagnose underlying performance issues more accurately. The model can help managers understand which variables most affect efficient and non-efficient processes (Entities), allowing for targeted improvement efforts.
- The *Human Performance Variation Prediction model* helps managers predict performance variability of consecutive work shifts by incorporating the human factor, supporting proactive workforce and resource planning aligned with human-centered, adaptable strategies.
- The *DMAIC-PDCA Structural Flowchart* equips managers with a structured, data-oriented representation of *ImproveXpert4.0*. It fosters organizational alignment and cross-functional collaboration with a real connection between top and operational management KPIs to promote globally sustained Continuous Improvement initiatives.
- As a unified solution, *ImproveXpert4.0* provides companies with an integrated platform that merges data from multiple sources, offering a comprehensive view of operations and enabling data-driven decisions. With continuous monitoring across production or service stages, companies can identify, e.g., quality issues early, leading to timely corrective actions. Automating data collection and analysis also frees up valuable resources, allowing employees to focus on strategic initiatives and innovation. Real-time insights enable companies to respond quickly to shifting business conditions or internal challenges, which is critical for staying competitive.

In addition, companies can implement more reliable and effective improvement measures by basing decisions on data rather than assumptions or intuition. The detailed process data analysis offered by ImproveXpert4.0 provides an opportunity to foster a genuine culture of Continuous Improvement.

5.3 Limitations and perspectives of future work

This study has two main limitations. First, while some participating companies, like Bosch, operate globally, the study's participants were exclusively based in Portugal, offering a localized perspective that may limit the findings' generalizability to international contexts due to differences in market conditions, cultures, and regulations. Second, while the algorithms developed are flexible and tailored to specific challenges, their broader applicability remains uncertain, as they have been tested only in two additional logistics cases and not across diverse industries.

In conclusion, future work from this study can focus on several key areas:

- *Application of the algorithms in other industries:* The algorithms developed should be analysed and applied to cases from other industries and business sectors. This will help prove their scalability and versatility across other contexts beyond the ones where they were tested;
- *Enhancement of the Human Performance Variation Prediction model:* The model's relevance would be significantly enriched by incorporating qualitative data collected directly from workers, offering a more comprehensive and human-centered perspective on their production challenges. Developing an approach to correlate these results with absenteeism patterns or predict potential health deterioration among workers would also represent a major step forward in advancing Industry 5.0 production systems focused on worker well-being and resilience.
- *Platform Development and Usability:* A preliminary prototype of the platform's graphical interface was created in collaboration with technology experts. However, further refinement is needed for a more visually appealing and user-friendly design. This interface must be then validated for usability by end users to ensure it meets practical requirements and expectations in its real-world setting usage;
- *User Training on the Algorithmic Modules and Problem Formulation:* Proposal to conduct training sessions in companies to use the algorithms integrated into the platform and to formulate new analysis problems based on their specific processes and available data;
- *Demonstration, Evaluation, and Communication:* To ensure rigor and transparency regarding the platform's practical relevance/novelty, the last three stages of the DSRM must be completed. First, the platform must be deployed and demonstrated in a company interested in testing and using it. Then, a performance evaluation should be conducted by comparing the company's results before and after using the platform and implementing its improvement plans. Finally, all gathered outcomes should be communicated/disseminated while respecting data confidentiality agreements.

Appendix A Interview Guide

1. How do Continuous Improvement projects appear in the company: mostly linked to the company's strategic vision, from top management metrics to the operational teams, or the reverse, in a bottom-top approach?
2. What methodology or set of steps and activities does the company follow to carry out Continuous Improvement projects?
3. How do you assess the results of implementing an improvement plan, i.e., how do you validate/analyse that the proposed improvement actions have produced beneficial results in process performance?
4. Which step(s) are the most challenging and time-consuming when conducting a Continuous Improvement project?
5. Thinking purely in terms of data-driven methodologies, techniques, or tools, which ones are used during the root cause analysis phase of the problems under analysis? - *Extra question put to participants P2, P4, P8 and P9.*
6. What types of data, and their sources, are collected and exploited by Continuous Improvement teams?
7. Has data on the individual performance of the company's employees already been used for Continuous Improvement projects? - *Extra question put to participants P4 to P10.*
8. How could emergent Industry 4.0 technologies enhance and expedite decision-making?

Author contributions

Conceptualization: Brochado, Â. F., Rocha, E. M., Pimentel, C.; Data curation: Brochado, Â. F.; Formal analysis: Brochado, Â. F.; Investigation: Brochado, Â. F.; Methodology: Brochado, Â. F., Rocha, E. M., Pimentel, C.; Software: Brochado, Â. F., Rocha, E. M.; Supervision: Rocha, E. M., Pimentel, C.; Validation: Brochado, Â. F., Rocha, E. M., Pimentel, C.; Visualization: Brochado, Â. F.; Writing – original draft: Brochado, Â. F.; Writing – review & editing: Brochado, Â. F., Rocha, E. M., Pimentel, C.

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Data availability

The data used to develop and test the algorithms presented in this work were provided by Bosch Thermotechnology, Vista Alegre S.A., Develop, and the Port of Sines. These organizations have not granted permission to share the data publicly. Additionally, the detailed information collected during expert interviews is not publicly available in order to protect individual privacy in compliance with the European General Data Protection Regulation.

Declarations

Ethics, consent to participate, and consent to publish

Not applicable.

Clinical trial

Not applicable.

Competing interest

The authors declare no conflict of interest.

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